Medical PMS & Billing Switch SDLC

Software Development Life Cycle & Ecosystem Blueprint for South Africa

Region: South Africa | **Standards: POPIA / CMS / HPCSA** | **Stack: Medplum / OpenEMR / X12 / Microservices**

1. EXECUTIVE SUMMARY & ECOSYSTEM CONTEXT

Developing a South African-compliant **Practice Management System (PMS)** with an integrated **Medical Billing Switch** requires uniting open-source clinical/FHIR standards with specialized South African electronic data interchange (EDI) clearinghouse protocols.

Primary Regulatory Bodies

- **CMS:** Council for Medical Schemes (ICD-10 clinical compliance).
- **HPCSA:** Health Professions Council of SA (Audit trailing & practitioner standards).
- **POPIA:** Protection of Personal Information Act (Data privacy & AES-256 encryption).

Key SA Medical Code Masters

- **ICD-10:** Mandatory primary & secondary diagnostic coding.
- **SAMA / Tariff Codes:** Procedural billing tariffs.
- **NAPPI Codes:** Surgical & pharmaceutical product mapping (MediKredit).
- **BHF Practice No:** 7-digit practitioner clearing identifier.

2. TECHNICAL ARCHITECTURE & BILLING SWITCH FLOW

The system utilizes a modern decoupled architecture where the open-source PMS communicates with a custom SA Billing Switch Engine via REST/JSON APIs, which then handles switching house protocol translation.

> **Architectural Flow Overview**
>
> **[Open-Source PMS]** (Medplum / OpenEMR) → *Generates Claim Payload* → **[Custom SA Billing Switch]** (Pre-Validation & Mapping) → *TLS 1.3 / HTTPS* → **[Switching Networks]** (MediSwitch / Healthbridge) → **[SA Medical Schemes]** (Discovery, GEMS, Bonitas)

Layer	Technology Options	South African Specific Role
PMS / EHR Core	Medplum (TypeScript/Node.js) or OpenEMR (PHP)	Captures patient demographics, 13-digit SA ID, medical scheme options, and clinical encounters.
Pre-Validation Engine	Node.js / Python / Go Microservice	Validates primary ICD-10 presence, NAPPI formatting, BHF number active status, and tariff calculations.
Switch Bridge	Custom Middleware (Docker / Code Engine)	

Layer	Technology Options	South African Specific Role
		Translates JSON/FHIR payloads into proprietary XML/Socket protocols required by MediSwitch & Healthbridge.
Infrastructure	IBM Cloud Code Engine & Cloudant / Postgres	Provides HIPAA & POPIA compliant encrypted storage (AES-256) and serverless execution.

3. 26-WEEK SOFTWARE DEVELOPMENT LIFE CYCLE (SDLC)

Phase 1 (W1-4) **Discovery, Regulatory & Architecture**

- Establish POPIA compliance framework, audit logging specifications, and encryption standards.
- Acquire master data files for SA ICD-10 tables, SAMA tariffs, and NAPPI code datasets.
- Apply for developer partnership access with SA switching networks (Altron / MediSwitch, Healthbridge).

Phase 2 (W5-8) **Core System Architecture & Schema Setup**

- Stand up open-source base (Medplum Node.js stack or OpenEMR) with extended SA demographic fields.
- Design microservice data models for the pre-validation billing switch engine.
- Set up containerized dev/staging/prod deployment environments on IBM Cloud Code Engine.

Phase 3 (W9-20) **Agile Sprint Execution (Sprints 1 - 10)**

Sprint	Module Focus	Key Deliverables
Sprints 1–2	Patient Management & Master Data	Patient intake with SA ID validation; ICD-10, Tariff, and NAPPI search/autocomplete.
Sprints 3–4	Encounter & Charge Capture	Clinical note workflows; linking ICD-10 diagnoses with SAMA/CPT procedural charges.
Sprints 5–6	Pre-Validation Engine	Automated checks for missing BHF, unmapped NAPPI codes, and rules-based price checks.
Sprints 7–8	Switch API Adapter	Direct HTTPS/Socket integration with MediSwitch/ Healthbridge; real-time response parsing.
Sprints 9–10	Remittance & Reconciliation	Processing batch Electronic Remittance Advices (ERAs) for auto-reconciling medical scheme payments.

Phase 4 (W21-24) **Testing & Switching House Accreditation**

- Execute end-to-end sandbox testing against switching house test environments (reversals, split-bills, co-pays).
- Conduct POPIA penetration testing, vulnerability assessments, and RBAC security audits.
- User Acceptance Testing (UAT) with SA practice managers, medical billing clerks, and clinicians.

Phase 5-6 (W25+)

Deployment, Onboarding & Continuous Maintenance

- **Phase 5 Go-Live:** Deploy production microservices; configure live practice BHF keys and routing credentials.
- **Phase 6 Operations:** Annual code updates (CMS ICD-10 tables, SAMA rates) and automated switch error monitoring.

4. COMPLIANCE & OPERATIONAL GUIDELINES

Key Quality Gate Checklist

POPIA Encrypted	Data encrypted at rest (AES-256) and in transit (TLS 1.3).
HPCSA Audit Log	Immutable logging of all clinician edits, patient views, and claim submissions.
CMS ICD-10 Validated	Automated rejection of non-standard clinical diagnosis codes prior to transmission.